costs. For example, a security camera can run a machine learning model to detect defects or safety violations on a production line and transmit only the alerts and metadata rather than streaming hours of raw video footage to the cloud.

In this book, we will look at how you can use some of the same cloud services that industrial organizations use today to help you build a digital twin.

### 1.2.3 Artificial intelligence (AI) and machine learning (ML)

The massive datasets from IoT sensors must be filtered, stored, indexed, and analyzed to generate actionable insights. Recent advances in AI and ML, partly enabled by specialized GPU hardware, have shifted these technologies into widespread use. Combined with IoT and cloud computing, they are essential for modern digital twins.

AI/ML capabilities are crucial for analyzing continuous sensor streams and making predictions about future system behavior. Cloud providers offer user-friendly tools that make these advanced capabilities accessible. Automated platforms handle the complex process of building, training, and deploying models—pre-processing sensor data, selecting algorithms, and optimizing performance without requiring deep technical knowledge.

For specific industries, specialized platforms offer pre-built solutions. When real-time responses are critical or cloud connectivity is unreliable, edge computing services allow AI processing to happen locally, enabling immediate decision-making. These tools collectively lower the barrier for organizations to add intelligent prediction, anomaly detection, and automated decision-making capabilities to their digital twins.

Furthermore, these capabilities are not only in the cloud. We will look at how you can download pre-trained models for common tasks such as object-detection, and timeseries forecasting, that can be customized and run locally, giving you both control over your data and the ability to operate when internet connectivity is unavailable.